AI and Data Scientist

Related Roadmaps

- Data Analyst Roadmap
- Prompt Engineering Roadmap

Find the detailed version of this roadmap along with other similar roadmaps

roadmap.sh

1 Mathematics

- [ ] Linear Algebra, Calculus, Mathematical Analysis
  - Mathematics for Machine Learning — Courses
- [ ] Differential Calculus
  - Coursera: Algebra and Differential Calculus — Course

2 Statistics

- [ ] Statistics, CLT
  - Coursera: Introduction to Statistics — Course
- [ ] Hypothesis Testing
  - Coursera: Hypothesis Testing — Course
- [ ] Probability and Sampling
  - Coursera: Probability and Statistics — Course
- [ ] AB Testing
  - Practitioner's Guide to Statistical Tests — Article
  - Experiment Design Article — Article
- [ ] Increasing Test Sensitivity
  - Minimum Detectable Effect — Article
  - Paper: Improving Test Sensitivity — Paper
  - Paper: Improving Sensitivity (CUPED) — Paper
  - CUPED at Booking.com — Article
  - Doordash: CUPAC — Article
  - Netflix: Stratification — Paper
- [ ] Ratio Metrics
  - Microsoft: Delta Method in Metric Analytics — Paper
  - Paper: Ratio Metrics — Paper

3 Econometrics

- [ ] Pre-requisites of Econometrics
  - Fundamentals of Econometrics — Book
- [ ] Regression, Timeseries, Fitting Distributions
  - Intro to Econometrics — Book
  - Coursera: Econometrics — Course
  - Kaggle: Learn Time Series — Course
  - Kaggle: Time Series Basics — Tutorial
  - ARIMA model for Time Series — Tutorial
  - Time Series Models — Tutorial
  - Forecasting Task with Solution — OpenSource
  - Coursera: Linear Regression — Course

4 Coding

- [ ] Learn Python Programming Language
  - Learn Python: Kaggle — Course
  - Google's Python Class — Course
- [ ] Data Structures and Algorithms (Python)
  - Algorithmic Excercises — Tutorial + Challenges
  - Study Plans - Leetcode — Challenges
  - Algorithms Specialization — Course
- [ ] Learn SQL
  - SQL Tutorial — Course

5 Exploratory Data Analysis

- [ ] Data understanding, Data Analysis and Visualization
  - Exploratory Data Analysis with Python and Pandas — Course
  - Exploratory Data Analysis for Machine Learning — Course
  - Exploratory Data Analysis with Seaborn — Course

6 Machine Learning

- [ ] Classic ML (Sup., Unsup.), Advanced ML (Ensembles, NNs)
  - Open Machine Learning Course - Open Data Science — Course
  - Machine Learning Specialization — Course
  - Pattern Recognition & ML by Christopher m. Bishop — eBook
  - GitHub repository with notes & code from the eBook above

7 Deep Learning

- [ ] Fully Connected, CNN, RNN, LSTM, Transformers, TL
  - Deep Learning Specialization — Courses
  - Deep Learning Book — eBook
  - Attention is all you need — Paper
  - The Illustrated Transformer — Article

8 MLOps

- [ ] Deployment Models, CI/CD
  - MLOps Specialization — Courses

Keep Learning